

Kanban Studio

Comprehensive Project Tutorial, System Architecture & Technical Documentation

1. Executive Overview & Project Description

Kanban Studio is an AI-powered, real-time multi-project task management system designed to emulate modern enterprise workflows. Built with a high-performance **FastAPI** backend, **SQLite** embedded database (WAL Mode), and a **Next.js / React 19** frontend, it enables teams to manage task cards across custom workflow columns with instant drag-and-drop mechanics, role-based authorization (RBAC), live WebSockets synchronization, due-date alerting, and analytical AI assistance.

2. Technology Stack & Frameworks

Layer	Technologies Used	Key Purpose & Benefits
Frontend Framework	React 19 / Next.js 16 (App Router)	Server-side static rendering, fast client hydration, and clean component hierarchy.
Styling & UI	Tailwind CSS v4, Glassmorphism Tokens	Curated color system for Light & Dark modes, WCAG AA contrast, responsive flex layouts.
Drag & Drop Engine	@dnd-kit/core, @dnd-kit/sortable	Smooth hardware-accelerated drag mechanics with full keyboard accessibility.
Backend Framework	Python 3.13 / FastAPI	Asynchronous RESTful APIs, OpenAPI auto-docs, Pydantic type safety.
Database	SQLite3 (WAL Mode & Compound Indexes)	ACID-compliant relational persistence, zero-config deployment, optimized query speeds.
Real-Time Engine	FastAPI WebSockets (ws_manager)	Instant state broadcasts to concurrent clients subscribing to project channels.
AI Engine	OpenRouter / Gemini 4o-mini / Local NLP	Natural language card mutations, project summary reports, workload & overdue analysis.
Testing & Audit	Pytest, Vitest, Playwright E2E	100% automated test coverage across unit, API, component, and user workflow levels.
Containerization	Docker, Multi-stage Docker Compose	Self-contained production deployment bundling static assets and Python runtime.

3. Complete Feature Breakdown (Parts 1 – 25)

- **Core Kanban & Card Drag-and-Drop:** Five workflow columns (Backlog, Discovery, In Progress, Review, Done) with smooth dnd-kit drag reordering and position persistence.

- **Rich Task Metadata Editor:** Edit card title, details, long descriptions, priority levels (High/Medium/Low), due dates, assignees (@username), and tags (#tag).
- **Filter, Search & Sort Toolbar:** Search cards instantly by title or details, filter by priority or assignee, sort by date/priority, and reset active filters with one click.
- **History Undo & Redo (Ctrl+Z / Ctrl+Y):** Optimistic state management storing up to 20 historical snapshots. Supports both hotkeys and toolbar UI buttons.
- **Multi-Project Workspaces:** Create, switch, and delete independent project workspaces with isolated board configurations and permissions.
- **Role-Based Access Control (RBAC):** Hierarchical authorization (Owner > Admin > Member > Viewer) protecting board mutations, invitations, and deletions.
- **Audit Activity Log:** Timestamped event tracking logging card creations, moves, updates, deletions, and member invitations per project.
- **WebSockets Real-Time Sync:** Live bi-directional WebSocket connection (/ws/projects/{project_id}) instantly syncing changes across active browser sessions.
- **Notification Center & Due-Date Alerting:** Automatic background scanner generating alerts for tasks due within 48 hours and drawer notification manager.
- **AI Project Intelligence:** Analytical quick presets: Project Summary, Workload Analysis, Overdue Task Identification, and Re-Prioritization Suggestions.
- **Production Hardening & Security:** IP-based rate limiting on authentication routes (max 15/min), global exception masking, and secrets isolation.
- **Performance Optimizations:** React component memoization (KanbanCard, KanbanColumn), SQLite compound indexing, and paginated API endpoints.

4. Step-by-Step User Tutorial & Guide

Step 1: Account Access & Login

Open <http://127.0.0.1:8000>. Login with username 'user' or register a new user. Default workspace boards auto-seed upon first sign-in.

Step 2: Workspace & Project Navigation

Use the top header dropdown to switch between projects or click '■ New Project' to create an independent workspace.

Step 3: Creating and Dragging Cards

Click '+ Add Card' in any column. Enter task title and details. Drag cards between columns to change state or reorder within a column.

Step 4: Editing Metadata & Assignees

Click any card to open the Edit Task Modal. Set priority (High/Medium/Low), due date, assignee, and tags.

Step 5: Inviting Team Members & Setting Roles

Click 'Members' in the header. Enter a team member's username and assign their role (Owner, Admin, Member, Viewer).

Step 6: Using AI Assistant & Intelligence

Click the 'AI Assistant' widget in the bottom-right corner. Use quick buttons like 'Project Summary' or 'Workload' for instant AI analysis.

Step 7: Checking History & Alerts

Click 'History' to review project activity logs or click 'Alerts' to check upcoming due-date notifications.

5. Database Schema & Architecture

The relational database schema is stored in **SQLite3** using WAL (Write-Ahead Logging) mode and compound indexes for high concurrency:

Table	Primary Columns	Indexes & Constraints
users	id, username, password_hash, created_at	PRIMARY KEY(id), UNIQUE(username)
boards	id, user_id, title, owner, created_at	PRIMARY KEY(id), FOREIGN KEY(user_id)
columns	id, board_id, title, position	PRIMARY KEY(id), FOREIGN KEY(board_id)
cards	id, column_id, title, details, priority, due_date, tags, assignee, position	PRIMARY KEY(id), FOREIGN KEY(column_id), idx_cards_due_date
project_members	id, project_id, user_id, role, created_at	UNIQUE(project_id, user_id), idx_project_members_user_project
activity_log	id, project_id, user_id, action_type, message, details, created_at	idx_activity_log_project_created
notifications	id, user_id, project_id, type, title, message, is_read, created_at	idx_notifications_user_read

6. REST & WebSocket API Endpoint Reference

Method	Endpoint Path	Description & Function
POST	/api/auth/login	User authentication with rate limiting protection.
POST	/api/auth/register	User registration & password hashing.
GET	/api/board	Fetch project board structure (columns & cards).
PUT	/api/board	Batch update board state (drag reordering).
GET/POST	/api/projects	List user projects or create new workspace project.
GET/POST	/api/projects/{id}/members	List or invite project team members (RBAC enforced).
POST	/api/ai/chat	Process AI assistant commands & project intelligence.
GET	/api/activity-log	Fetch paginated audit activity history log.
GET	/api/notifications	Fetch paginated user notifications & due date alerts.
WS	/ws/projects/{id}	Real-time WebSocket project change broadcast channel.

7. Local Setup & Docker Deployment Guide

Run the production stack locally using Docker Compose:

```
# Build and launch production container stack docker-compose up --build -d # Verify backend health check curl http://localhost:8000/api/health
```

8. Conclusion & Verification Summary

Kanban Studio is fully validated with **22 Pytest backend tests**, **44 Vitest frontend unit tests**, and **14 Playwright end-to-end workflow tests**. All 25 specification parts have been completed and verified for production readiness.